

# Cloud Computing & DevOps – AWS Handbook

## SESSION 7

### Docker Fundamentals

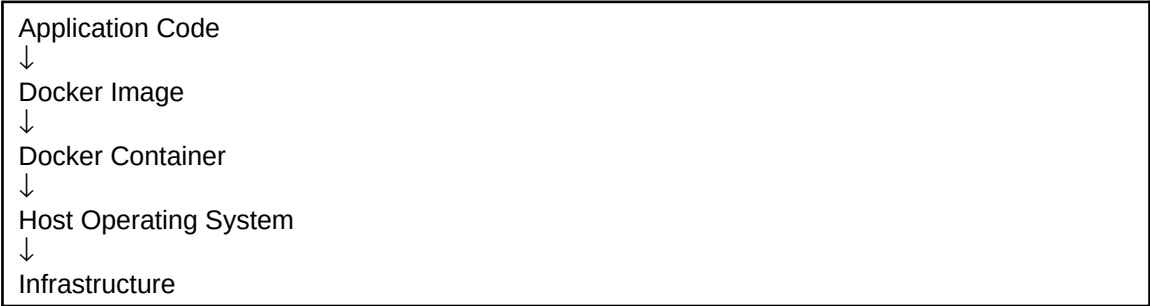
#### Introduction to Containerization

|                         |                                                                                                      |
|-------------------------|------------------------------------------------------------------------------------------------------|
| <b>Duration</b>         | 2 Hours                                                                                              |
| <b>Prerequisites</b>    | Linux Fundamentals, EC2 Knowledge                                                                    |
| <b>Learning Outcome</b> | Understand containers, images, Docker architecture, and deploy applications using Docker containers. |

# Why Are We Learning This?

Modern applications are increasingly deployed using containers instead of traditional virtual machines. Containers provide portability, consistency, faster deployment, and efficient resource utilization. Docker is the most widely adopted containerization platform and forms the foundation of modern DevOps practices.

## Architecture Diagram



## Key Concepts

|                    |                                             |
|--------------------|---------------------------------------------|
| Container          | Lightweight isolated execution environment. |
| Docker Engine      | Core service that manages containers.       |
| Docker Image       | Blueprint used to create containers.        |
| Docker Container   | Running instance of an image.               |
| Dockerfile         | File containing image build instructions.   |
| Container Registry | Repository used to store images.            |

## Practical Activities

- Install and verify Docker
- Run first Docker container
- Explore Docker images
- Create a Dockerfile
- Build a custom Docker image
- Deploy a web application inside a container
- Access the application through a browser

## Expected Outputs

Docker installed successfully, containers running, custom image created, and application accessible through a web browser.

## What You Should See

**Paste Screenshot Here**

Docker Version Verification

**Paste Screenshot Here**

Docker Images List

**Paste Screenshot Here**

Running Containers

**Paste Screenshot Here**

Docker Image Build Output

**Paste Screenshot Here**

Application Running in Browser

# Journal Writing Template

## Aim

To understand Docker containerization and deploy an application inside a Docker container.

## Theory

Docker provides containerization technology that packages applications along with their dependencies.

## Procedure

Write 5–7 concise steps performed during the practical session.

## Observations

Docker was installed successfully, images were created, and containers executed correctly.

## Conclusion

Docker simplified application deployment by packaging applications into portable containers.

# Viva Preparation

1. What is Docker?
2. What is Containerization?
3. Difference between Containers and Virtual Machines?
4. What is a Docker Image?
5. What is a Docker Container?
6. What is a Dockerfile?
7. Why is Docker important in DevOps?

## Troubleshooting Box

| Problem                 | Possible Cause                   |
|-------------------------|----------------------------------|
| Container not starting  | Incorrect image or configuration |
| Port not accessible     | Port mapping issue               |
| Image build failed      | Dockerfile error                 |
| Application unavailable | Application service not running  |

## Industry Relevance

Docker is the standard container platform used across cloud providers and enterprise environments. It enables consistent deployment from development to production and is a prerequisite for Kubernetes and modern DevOps workflows.

## Session Summary

Students learned Docker fundamentals, containerization concepts, image creation, container deployment, and application hosting using Docker.